

Milestone Two Narrative

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Briefly describe the artifact. What is it? When was it created?

The artifact I selected for enhancement is a full-stack, data-driven web application dashboard designed for Grazioso Salvare, an international animal rescue organization. I created this during my CS 340 Client-Server Development course. This application uses Python Dash and Plotly to render interactive filtering tables, demographic pie charts, and real-time geospatial coordinate-tracking maps. The system allows rescue personnel to dynamically sort and filter specialized animal populations, such as water-rescue or disaster-tracking dogs, providing crucial data visualizations to support critical organizational decision-making.

Justify the inclusion of the artifact in your ePortfolio. Why did you select this item? What specific components of the artifact showcase your skills and abilities in software development? How was the artifact improved?

I included this artifact in my ePortfolio because it represents a complete, end-to-end full-stack computing solution and demonstrates a mature understanding of client-server communication and interface interactions. To showcase my advanced software development skills, the original codebase was overhauled and improved in three specific areas:

- **Architectural Separation of Concerns:** The loose, procedural script layout of my original project was refactored into a strict, module-driven Object-Oriented Programming (OOP) design. By isolating the front-end views (Dashboard.py) and encapsulating data interactions into a standalone database access class (AnimalShelter.py), my software is now decoupled, making it production-grade, reusable, and easy to maintain.
- **Defensive Programming Elements:** To elevate my software robustness, strict data-validation traps and conditional emptiness guards were engineered directly into the

callbacks. This prevents the application from throwing sudden runtime crashes when input fields are unpopulated or malformed, demonstrating a commitment to reliable software quality.

Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

The technical enhancements I implemented successfully met the course outcomes originally targeted in Module One:

Outcome 3: Design and evaluate computing solutions... managing design trade-offs. I achieved this by refactoring the procedural layout into decoupled OOP files, deliberately managing the structural trade-off between basic, single-file scripts and highly maintainable, modular enterprise software architectures.

Outcome 4: Demonstrate an ability to use well-founded techniques, skills, and tools. I was able to demonstrate this by implementing clean, class-based import modules, virtual environment isolation protocols via VS Code, and migrating the application engine from browser-dependent notebooks to a native, standalone Dash compilation framework.

Outcome 5: Develop a security mindset... to mitigate design flaws. This was achieved by constructing server-side error traps and frontend boundary checks to intercept empty datasets, along with a secure configuration layer that separates environment variables from the core execution logic.

Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

During the process of modifying and improving my artifact gave me valuable insight into software lifecycle maintenance. I learned that refactoring legacy code requires an equal focus on system backward compatibility and structural optimization. I ran into some challenges while developing the environment configuration layers as I moved the application from an interactive Jupyter notebook to a standalone native web application in Visual Studio Code. Overcoming this roadblock involved debugging package dependency collisions and shifting obsolete runtime methods over to modern, standard Dash compilation structures. Resolving these challenges taught me how to trace system failures and execute defensive coding practices, building significant confidence in my capabilities as a professional software engineer.